

# Global Office

## Architecture Design Document

The engineering blueprint of an AI-native operating system for organizations: a graph-native data substrate, a multi-agent cognition layer, and a database-enforced multi-tenant runtime deployed at the edge.

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|-----------------------|------------------------------------------------------------|
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| <b>DOCUMENT OWNER</b> | Office of the CTO / Engineering                            |
| <b>APPLIES TO</b>     | globaloffice.cloud production estate                       |
| <b>VERSION</b>        | 1.0 — August 2026                                          |

# Executive Summary

Global Office is not a suite of applications with an AI feature bolted on. It is a single operating substrate in which every organizational event — a task, a document, a message, a meeting, a KPI movement — is written once into a tenant-scoped knowledge graph, and every intelligent behaviour is derived from that graph. The consequence is architectural, not cosmetic: the intelligence improves as the organization operates, and it cannot be replicated by a competitor stitching an LLM onto a CRUD product.

## Five design commitments

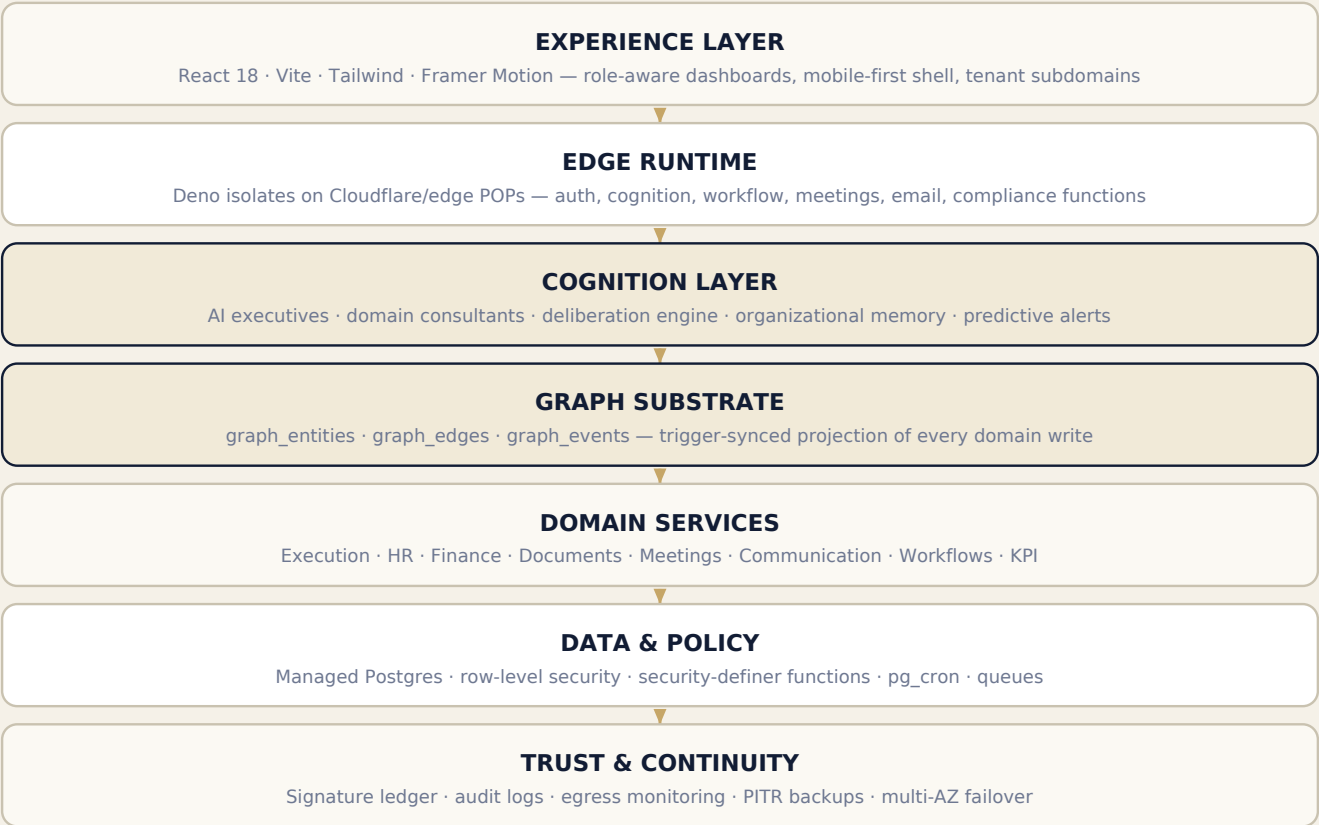
- **Graph-native by default**  
Domain writes are mirrored into `graph_entities` / `graph_edges` by database triggers, so no feature can ship without contributing to organizational memory.
- **Isolation enforced in the database**  
Tenancy is a row-level security property of Postgres, not application middleware. A bug in UI code cannot leak another tenant's data.
- **Cognition as a governed pipeline**  
Multi-agent deliberation is recorded step-by-step, policy-bounded, human-approved, and replayable — advisory by design.
- **Stateless edge compute**  
All server logic runs as isolated Deno functions at the edge; horizontal scale is a configuration value, not a re-architecture.
- **Evidence over assertion**  
Audit ledgers, signature chains, and access logs are first-class tables, which is what makes compliance claims demonstrable.

## System at a glance

|                |                |                 |              |          |               |
|----------------|----------------|-----------------|--------------|----------|---------------|
| 7              | 18+            | 79              | 100%         | <300ms   | 99.9%         |
| runtime layers | edge functions | governed tables | RLS coverage | p95 read | uptime target |

# Layered Architecture

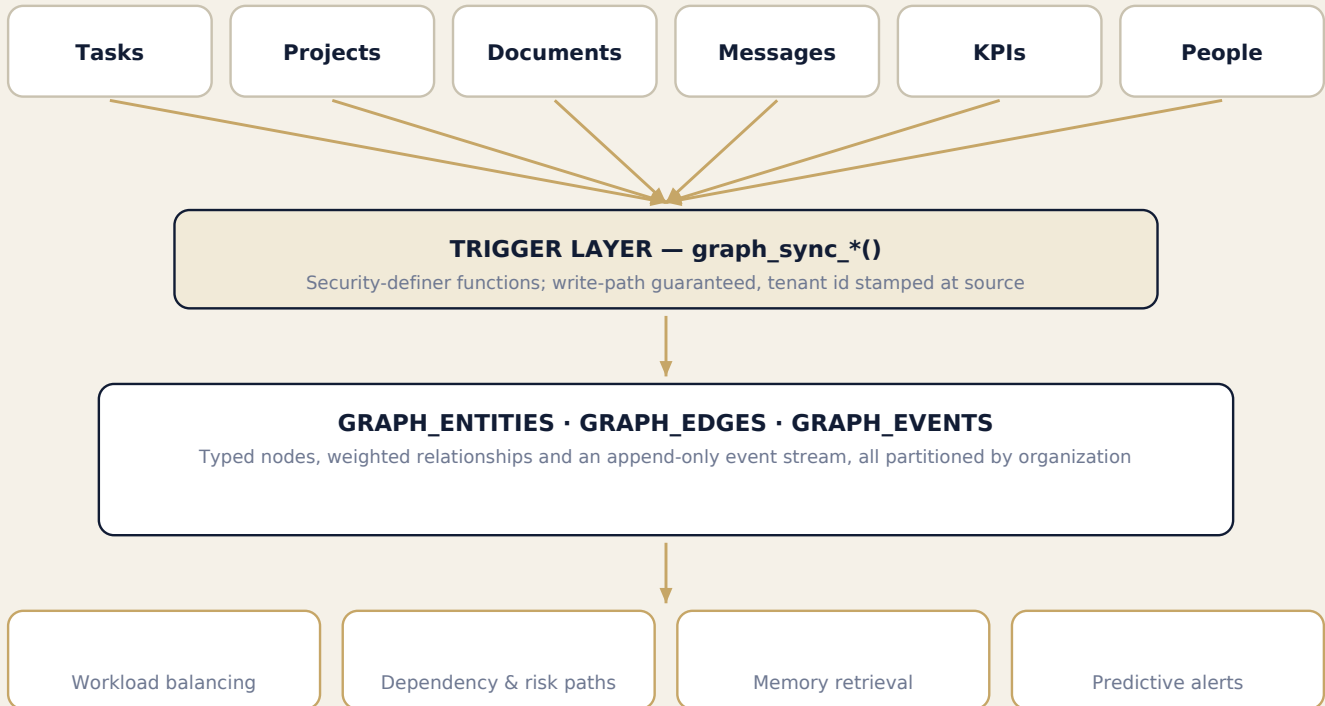
Seven layers, each replaceable in isolation. Requests enter at the edge, are authorized in the database, and every mutation emits a graph event that feeds the cognition layer.



**Layers 3 and 4 (highlighted) are the defensible core. Competitors can copy layers 1, 2 and 5 in a quarter; they cannot retroactively populate a customer's graph or its accumulated deliberation history.**

# The Graph Substrate

Every domain table has a database trigger that upserts a canonical entity and its edges. The graph is therefore never stale, never optional, and never dependent on application code remembering to write it.



## Why this matters commercially

Switching cost is not contractual, it is epistemic. A customer leaving Global Office abandons the machine-readable record of how their organization actually works — who depends on whom, which decisions were taken and why, which risks were flagged and how they resolved. That record has no export format in any competing product.

# The Cognition Pipeline

Requests for judgement enter a governed pipeline. Each stage is persisted, so any recommendation can be replayed and audited months later — the property enterprise buyers actually require before trusting AI with operations.



## Advisory by construction

No agent holds write authority over payroll, contracts, personnel or funds. Autonomy is a per-tenant policy row, defaulting to suggest-only. This is a deliberate architectural constraint: it keeps the platform on the safe side of emerging AI governance regimes while still delivering the decision quality of an executive team.

# Tenancy, Identity & Security

Every tenant receives its own subdomain on globaloffice.cloud and, more importantly, its own enforced data boundary inside a shared, cost-efficient database.



## Controls in force

|                 |                                                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Data at rest    | AES-256 on managed Postgres and object storage                                                                                    |
| Data in transit | TLS 1.3 everywhere, HSTS on all tenant subdomains                                                                                 |
| Privilege model | Six roles, separate user_roles table — role escalation structurally impossible from the client                                    |
| Membership      | Emailed invitation or single-use, role-scoped, expiry-bounded access token; owner never grantable by token                        |
| Secrets         | Vault-managed, never shipped to the browser, rotated on schedule                                                                  |
| Data room       | Two-tier: approved investors see cleared documents; admin-only material is filtered by policy in both database and object storage |
| Recovery        | Point-in-time recovery, multi-AZ replicas, RPO 5 min / RTO 60 min                                                                 |
| Evidence        | signature_ledger, inter_org_audit_log, access_token_events, egress_events                                                         |

# Scale, Performance & Cost Architecture

The architecture was chosen so that unit cost falls as usage rises. Compute is stateless and elastic, storage is tiered, and inference is routed by task complexity rather than defaulting to the most expensive model.

## Cost-control mechanisms engineered into the platform

- **Model-tier routing**  
Cheap models answer routine questions; frontier models are reserved for deliberation. Cost per intelligent action falls without touching price per seat.
- **Retrieval caching**  
Graph and memory lookups are cached per tenant, removing repeat inference on stable context.
- **Metered AI credits**  
Consumption above tier allowance is billed, so the heaviest users fund their own inference and expansion revenue arrives at near-full contribution margin.
- **Shared multi-tenant substrate**  
One elastic database estate serves all tenants with isolation enforced logically — no per-customer infrastructure to provision or pay for.
- **Scheduled work off the hot path**  
pg\_cron and queues absorb alerts, summarization, digests and cleanup, keeping p95 read latency flat under load.

## Operating envelope

| DIMENSION             | TARGET           | HOW IT IS ACHIEVED                                   |
|-----------------------|------------------|------------------------------------------------------|
| Read latency (p95)    | <300 ms          | Edge execution, regional caching                     |
| Availability target   | 99.9%            | Multi-AZ managed Postgres + edge runtime             |
| Deployment            | <5 minutes       | Continuous delivery, zero-downtime releases          |
| Scale path            | Horizontal       | Stateless isolates, read replicas, partitioned graph |
| Cost per organization | Falls with scale | Committed-use discounts, caching, model routing      |

# Architectural Roadmap

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The expensive half of the engineering journey is complete. The roadmap below extends the same substrate rather than replacing it — there is no rewrite on the horizon.

## SHIPPED

Graph substrate, cognition pipeline, workflow engine, meetings intelligence, trust ledgers, multi-tenant subdomains, investor data room

## NEXT 6 MONTHS

Enterprise SSO/SAML depth, SLA tiering, regional data residency, expanded connector surface

## 6-18 MONTHS

Marketplace for third-party modules, Partner Connect network graph, tenant-tuned model routing

## 18 MONTHS +

Cross-organization benchmarking on privacy-preserving aggregates; the intelligence layer as an ecosystem platform

## Closing note for reviewers

Read this document as an argument about durability. Products are copied; substrates are not. Global Office writes every organizational event into a structure that becomes more valuable to the customer and more expensive to abandon with each passing week of use, while the cost of serving that customer declines. That is the whole design, and every layer above exists to serve it.